

## Video Transcript

### Video 1: Introduction (03:55)

Java is something we haven't addressed yet. In many ways we have tried to avoid it. It is a language that imposes a lot of structure on you. It is a very different environment. It is mostly in legacy systems at this point and so we have not addressed it. However, when it comes to data platforms, especially those from the Apache Software Foundation, there's a great deal of Java there and there are many projects that are written in Java. Just to list a few, Spark, Kafka, Cassandra, Hadoop, Elasticsearch, HDFS, CouchDB and many others. And so because of it, it is good to have some basics about the language, some basics, and to how to set up an environment to run in. And so, that's what we will be doing in this section. This will be a mini crash course on Java to be able to get some of those basics that I mentioned that would at least help you to run some of those projects that require a Java platform.

When it comes to doing change data capture, as you can imagine, doing it by hand, it's not such a great idea. You can certainly do it, we did it, but it is more of an educational type of approach. We want to understand the issues. And although there are shops out there that maintain this type of systems by hand, there are better alternatives. It feels at times like a Rube Goldberg machine, where you have too many moving pieces and your risk of something breaking is too high. Now, as you can imagine, change data capture is a problem that is well recognized, and because of it there have been a number of projects that have dedicated efforts and come up with great solutions. Now, there's a number of these and we will sample these in this section. We'll start off with ones that are in Python and that are a great fit to where we have focused our efforts. They're light, they're easy, you bring them in, and you can monitor a system pretty well.

Now, after we do that, we will be looking at some of the really industrial solutions, meaning these scale at the highest level and they are written in platforms such as Java. Here, we will use containers once again to be able to forgo all of the installation that we would need to be able to get that going. But it's a great way to observe a full change data capture solution, that is, the one that is used at the highest level, meaning at the highest scale when it comes to data. So, now let's go ahead and jump in.

As we have seen, many of the data platforms are from the Apache Foundation. And if you were to take a closer look at the many, many projects that the Apache Foundation has, you'll notice that a lot of them are written in Java. And as you can see here, some of the major building blocks when it comes to data engineering projects like Spark, Kafka, Cassandra, they're all written in Java. Now, there's something to say about Java in that although it is younger than Python, you can see there the dates, it has a lot more structure. And because of it, it is not one of the most loved languages.

As you can see there, it is certainly not in the top half. Now, more than that, as you can see as well, it's also in the top half of the most dreaded languages. However, even though it has those shortcomings, it is widely used in the market. You can see there that it's at the top of popularity, meaning the projects and the number of developers that exist out in the market. And so because of it, we're going to take a look at some of the basics as you're bound to touch Java projects sooner or later when it comes to your data engineering career. And this is good knowledge to have.

